

Assessment and Treatment of Cutaneous Neurofibromas in Neurofibromatosis Type 1

A Scoping Review

Hadiya Abdalla Elahmar,¹ Carlos Alberto Soto Rincon,² Aaron Drucker,^{1,3,4} Patricia Parkin,^{1,5} and Carolina Barnett-Tapia^{1,2}

Correspondence

Dr. Elahmar
hadiya.elahmar@mail.utoronto.ca

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Abstract

Purpose of Review

Cutaneous neurofibromas (cNFs) are a hallmark feature of neurofibromatosis type 1 (NF1) and are associated with substantial physical, psychosocial, and quality-of-life burden. Progress in therapeutic development is hindered by the absence of validated, feasible, and responsive tools for quantifying cNFs and assessing their clinical effect and treatment response. This scoping review aimed to identify and summarize existing methods used to measure and quantify cNFs, evaluate their effect on health-related quality of life (HRQoL), and assess outcome measures used in cNF treatment studies.

Recent Findings

A comprehensive search of 6 electronic databases (MEDLINE, Embase, Cochrane Central Register of Controlled Trials, Web of Science, ClinicalTrials.gov, and the International Clinical Trials Registry Platform) from inception to April 2025 identified 1,916 records, of which 61 articles met inclusion criteria (48 primary studies and 13 review articles). Eight studies evaluated instruments for cNF measurement, while 5 assessed HRQoL using patient-reported outcomes (PROs). A greater cNF burden, particularly an increased number, size, and facial involvement, was consistently associated with poorer HRQoL. Among identified PROs, the cNF-Skinindex was the only tool specifically developed for cNFs. Twenty-three studies investigated therapeutic interventions, and 12 registered clinical trials evaluated medical or surgical treatments. Considerable heterogeneity was observed in outcome selection, with many studies lacking clearly defined end points or relying on nonvalidated or inconsistently applied measures.

Summary

This scoping review highlights the availability of several methods to quantify cNFs; however, their clinical feasibility, responsiveness, and interpretability remain limited. Outcome measures used across cNF intervention studies are highly variable, impeding comparison across studies and synthesis of evidence. The cNF-Skinindex represents a promising cNF-specific HRQoL instrument but requires further validation, including assessment of responsiveness and minimal important difference thresholds. Standardization of cNF measurement approaches and outcome selection is urgently needed to support robust clinical trials and enable meaningful evaluation of treatment efficacy in NF1.

Introduction

Cutaneous neurofibromas (cNFs) are a key feature of neurofibromatosis type 1 (NF1).¹ Their number and size generally increase with age, varying across body regions.^{2,3} Most cNFs appear on the trunk (60%), with about 10% on the head and neck.^{2,3} Their appearance ranges from flat, discolored lesions to large, pedunculated growths.⁴ These lesions often cause progressive

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Supplementary Material

¹Institute of Health Policy, Management and Evaluation, Dalla Lana School of Public Health, University of Toronto, Ontario, Canada; ²Elisabeth Raab Neurofibromatosis Clinic, Toronto General Hospital, University Health Network, Ontario, Canada; ³Department of Medicine and Research and Innovation Institute, Women's College Hospital, Toronto, Ontario, Canada; ⁴Division of Dermatology, Department of Medicine, University of Toronto, Ontario, Canada; and ⁵Department of Paediatrics, Faculty of Medicine, University of Toronto, Ontario, Canada.

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Glossary

cNF = cutaneous neurofibroma; **HRQoL** = health-related quality of life; **ICC** = Intraclass Correlation Coefficient; **MDD** = minimal detectable difference; **MEK** = Mitogen-Activated Protein Kinase Kinase; **MID** = minimal important difference; **NF 1** = neurofibromatosis type 1; **OCT** = optical coherence tomography; **PDT** = photodynamic therapy; **PRO** = patient-reported outcome; **PS** = photosensitizer; **QoLI** = QoL Index; **RCT** = randomized controlled trial; **SFDI** = spatial frequency domain imaging; **YAG** = Yttrium Aluminium Garnet.

disfigurement, discomfort, and functional limitation, prompting many patients with NF1 to seek medical care.^{3,5}

cNFs significantly affect quality of life (QoL) for people with NF1 across physical, emotional, and social domains. Their visibility often affects body image, self-esteem, and social interactions.⁶ Research highlights that factors such as facial involvement and total cNF count are linked to reduced QoL.^{7,8} As a result, clinical trials are increasingly incorporating QoL as a secondary outcome, using patient-reported outcomes (PROs) to capture the broader effects of treatment. Generic health-related QoL (HRQoL) measures may lack sensitivity to assess the effects of cNFs on QoL.

Standardized systematic and prospective evaluation of current treatment outcomes across different cNF subtypes, skin types, and age groups is necessary to improve efficacy assessment and guide future therapies. This underscores the importance of current initiatives, including those by the REINS cNF subgroup, which aim to develop standardized outcome measures and validated assessment tools to improve the consistency and comparability of cNF research.

The objective of this scoping review is to assess existing literature and identify gaps in measuring cNFs, their effect, and response to therapy.

Methodology

Databases and Search Strategy

We searched Medline, Embase, Cochrane Central Register, Web of Science, ClinicalTrials.gov, and International Clinical Trials Registry Platform from inception until April 2025. Languages were restricted to English, Spanish, and French. The search strategy was developed and refined with an experienced librarian (eMethods). All pooled citations were imported into a reference manager (EndNote 20) and then exported to Covidence, an abstract managing site. Duplicate studies were automatically removed by Covidence software, and any remaining duplicates were manually removed by reviewers.

Inclusion Criteria

We included retrospective and prospective observational studies, cross-sectional studies, case series ($n > 3$), and randomized controlled trials (RCTs), involving individuals of all ages and sexes diagnosed with NF1 according to established NIH diagnostic criteria.^{1,9} We included manuscripts specifically reporting the

following: (1) quantification of cNFs including size, volume, and number; (2) effect of assessment of cNFs in HRQoL; and (3) treatment options for cNFs (eAppendix, eTable 1).

Exclusion Criteria

Conference abstracts, case reports, reviews reporting on studies involving other genetic skin conditions, and those addressing NF1 manifestations other than cNFs.

Selection of Articles

Two reviewers (HAE, C.S.) independently screened all abstracts and titles to identify studies meeting the criteria for full-text review, and then independently and in duplicate screened all full texts to identify studies meeting inclusion criteria; any conflicts were arbitrated by a third reviewer (C.B.T.). The reference lists of included and relevant review articles were manually searched for additional references.

Data Extraction and Reporting

Both reviewers (H.A.E., C.S.) independently extracted data from the included studies into a standardized data extraction form on Microsoft Excel. The third reviewer (C.B.T.) was consulted to resolve disputes. The primary outcomes included quantitative and qualitative tools for assessing cNFs, available treatment options, categorized into medical and destructive approaches. Studies were evaluated based on the tools' intended purposes; when available, validated measurement properties of each instrument were extracted. Medical treatments were categorized into oral, topical, and systemic options, while destructive treatments were grouped into lasers, surgery, and electrodesiccation. Data were summarized by narrative synthesis as follows (eTables 2–4):

1. Studies on quantification methods for cNFs,
2. Studies on the effect of cNFs on HRQoL, and
3. Studies on therapeutic strategies for cNFs.

Data Availability

Data not provided in the article because of space limitations may be shared at the request of any qualified investigator for purposes of replicating procedures and results.

Results

A total of 1,924 records were identified through database and citation searching. After screening and eligibility assessment, 61 studies were included in the final review. The full study selection process is illustrated in the PRISMA diagram (Figure).

Studies on cNFs Quantification

We identified 9 published articles on cNF measurement from 2014 to 2024. Eight studies^{10–17} focused on device validation, and 1 was a clinical trial¹⁸ (summarized in eTable 5). These studies highlighted that cNFs are often numerous and heterogeneous, ranging from 1 to 30 mm in size, making measurement challenging and time-consuming. Quantification methods varied, including count, size (length, width, height), volume, surface area, and depth, using tools such as digital calipers,^{12,13,15,16} paper frames,^{10,12} two-dimensional (2D) and three-dimensional (3D) photography,^{13,15–17} spectral imaging,¹⁴ and high-frequency ultrasound sonography (HFUS).¹¹ Table 1 highlights the merits, disadvantages, and measurement properties of each tool.

Digital Calipers

Studies found calipers effective for measuring the length, width, and height of cNFs over 5 mm but unsuitable for flat or early lesions.^{12,13,15,16} Their moderate reliability (Intraclass Correlation

Coefficient [ICC] = 0.62–0.77)¹⁵ is lower than 3D photography (ICC = 0.93–0.98)^{13,15,17} and HFUS (ICC = 0.97–0.99),¹⁵ partly due to cNF pliability affecting pressure-sensitive readings. Calipers also lack durable source data for reassessment and are time-consuming, because each lesion must be measured manually. To improve feasibility, researchers often focus on a single body region, although this limits broader cNF evaluation.¹⁵

Two clinical trials used calipers to measure tumor size as a response end point for evaluating the efficacy and safety of mirdametinin¹⁹ and selumetinib²⁰ (Table 2).

2D and 3D Imaging

Five studies used 2D and 3D cameras, along with a stereoscopic optical imaging system, to assess cNFs.^{13,15–18} 2D imaging aids in localizing tumors and evaluating their color, number, and distribution across different body regions.

Figure PRISMA Diagram

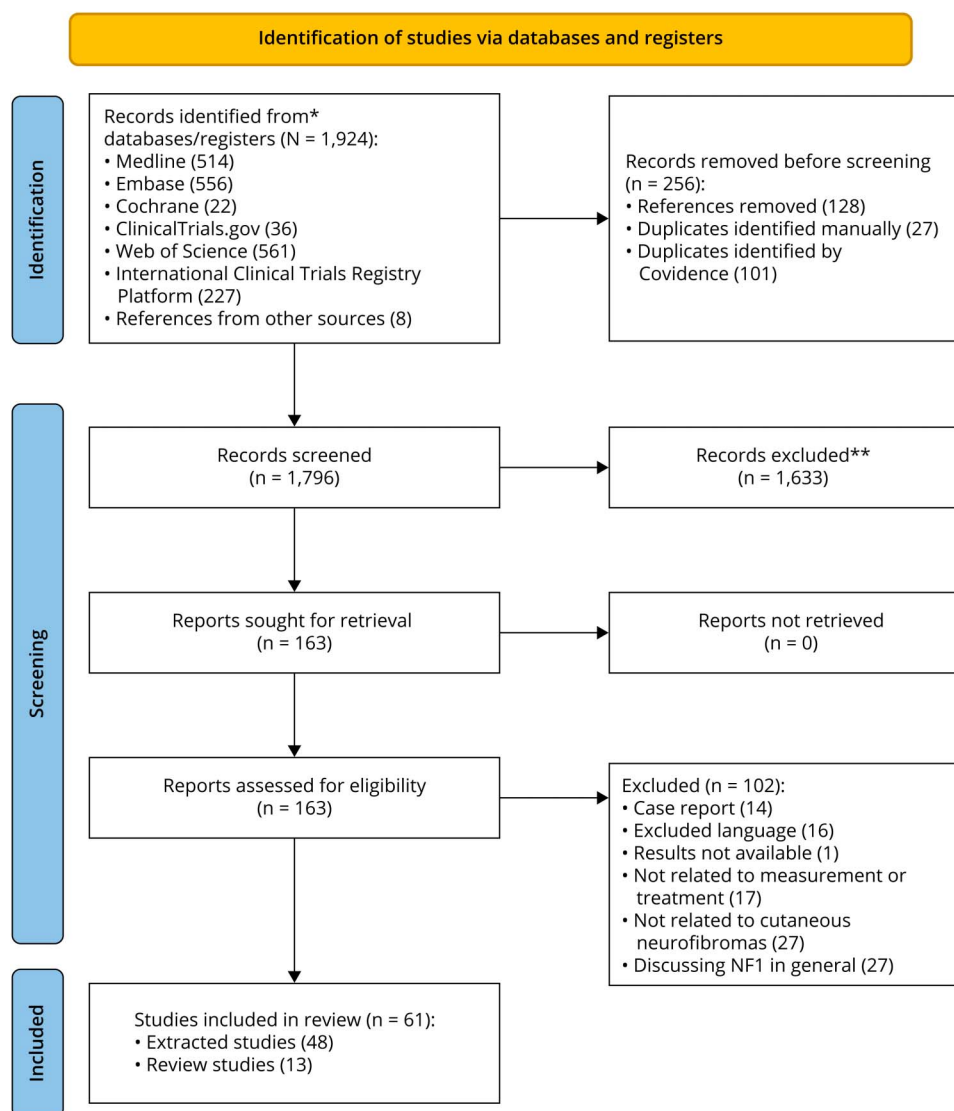


Table 1 Summary of Measures to Quantify cNFs

Technique	Notes	Advantages	Disadvantages	Reliability	Time required	Responsiveness	Cost
Digital calipers	Measure the longest diameter of large cNFs (>5 mm)	Easy to use	• Cannot measure depth below the skin surface • Does not create a permanent record • Low reliability for cNF height • Not suitable for cNFs <5 mm • Does not evaluate cNF morphology • Time-consuming if multiple cNFs	ICC = 0.62–0.77	6 minutes per 6 cNFs	Not assessed	\$50–500
2D, 3D photography	Best for measuring the surface area of large cNFs (>5 mm) and linear measure of small cNFs (<5 mm) Best for large areas of skin (10 × 10 cm)	No radiation. High accuracy supports real-time imaging and permanent recording	• Unable to measure subcutaneous cNF	ICC = 0.93–0.98	Three-dimensional imaging only requires 30 seconds	Not assessed	\$11,800–23,600
HFUS	Measure volume, width, and depth (above and below the skin) Assess skin layer position, surface characteristics (flat or protruding), margins, echogenicity, posterior acoustic enhancement, linear lateral extension, and plexiform pattern of cNFs	No radiation; high resolution	• Unable to measure cNF deep under the skin; sensitive to operator technique • Fairly expensive	ICC = 0.97–0.99	Ultrasound takes approximately 5 minutes	Not assessed	\$ 225,000
Spatial frequency domain imaging	Assesses skin's scattering and absorption, creating maps that can identify potential nascent cNFs by detecting optical differences from surrounding tissue	Allowing the detection of nascent cNFs	• Extensive training required for image acquisition and analysis • Technology is relatively brand new, and validation research is still ongoing	—	2 min for an area of 10 × 20 cm	Not assessed	\$63,000
Optical coherence tomography	High-resolution, three-dimensional OCT imaging enables detailed assessment of the skin's microstructure and microvascular alterations associated with cNF.	Permit cross-sectional view of lesions in vivo and visualization of vascular properties, microstructure, dermal-epidermal junction, and dermal layers	• Extensive training required for image acquisition and analysis	—	2 min for an area of 1 × 1 cm	Not assessed	\$87,000
Paper frame	A square measuring 100 cm ² was applied to 3 specific areas of the body (back, abdomen, and thighs)	Easy to use	• Time consuming	ICC = 0.992 to 0.997	Not provided	Not provided	—

Abbreviation: cNF = cutaneous neurofibroma; OCT = optical coherence tomography.

3D scanning captures cNFs in 3 dimensions with high precision, allowing accurate measurement of width, length, height, surface area, and volume. With proper training, operators can achieve high reliability (ICC ≥ 0.95).¹⁶ Handheld 3D imaging systems like Vectra H1, LifeViz Micro, and Cherry Imaging show comparable accuracy, with LifeViz being the fastest and most consistent. Vectra excels in measuring larger areas, while Cherry Imaging is best for

uneven surfaces.^{15,17} Whole-body (WB) cameras, although lower in resolution, track multiple cNFs over time but require more space and cannot detect subcutaneous tumors.¹⁵ An ongoing trial with Vectra WB360 shows that it can detect cNFs as small as 4 mm in individuals aged 1–69, although it may have volume measurement errors. The analysis of the cNF count through these digital photographs is still ongoing.²¹

Table 2 Studies Assessing Drug Therapies for cNFs

Treatment	Citation/ registration	Year	No of patients	Study design	Outcome measures	Mode of action
5% imiquimod cream	NCT00865644	2009	11	Prospective	Measure tumor volume (did not specify the used tool)	Toll-like receptor 7 agonist
Diclofenac topical	Oliveira et al.	2021	6	Prospective	Visual and dermatoscopy observations, measurements (measured in 2 axes using measuring tape), photographs, and histopathology	COX-1, COX-2
Diclofenac injections	Geller et al.	2015	7	Prospective	Measurement (the exact measured dimensions in 2 axes). Photography of both lesions (accompanied by a measuring tape)	COX-1, COX-2
Sirolimus	Kaneda et al.	2023	18	RCT	Photographs, CT images of a target cNF and ruler measurements	mTOR inhibitor
Everolimus	Slopis et al.	2018	22	Prospective	Photographs, volumetric and height measurement of lesions	mTOR inhibitor
Selumetinib	Gross et al. (NCT02839720)	2025	11	Prospective	Measured with calipers and volumes calculated by multiplying length, width, and height (reported responsiveness) Changes in skin-related morbidity assessed by Skindex 29	MEK 1/2inhibitor
Mirdametinib	NCT06159166	2023	24	Prospective	3D photography (sum of the surface area, sum of the tumor volume, sum of the longest diameter) and digital caliper (sum of the longest diameter)	MEK 1/2inhibitor
NFX-179 topical gel	NCT05005845 Sarin et al.	2023 2024	199 48	RCT RCT	Determine safety and effectiveness of 2 concentrations of NFX-179 gel Reduction of cNF volume, based on ruler measurements of longest diameter and height using a calculation of volume as a cylinder	MEK inhibitor
Ranibizumab	NCT00657202	2008	11	Prospective	Photographs and measurement of tumors by calipers	Antiangiogenesis
Photodynamic therapy (PDT)	Quirk et al. NCT02728388	2021 2016	4 30	Prospective	Change in lesion size (by measuring the long and short axis of the lesion) Tumors will be measured by digital photography and ultrasound	Photosensitizer taken up by tumor cells is activated by light to induce targeted tumor destruction
Diphencyprone (DPCP)	NCT05438290	2022	17	Prospective	Photographs, along with whole-body photography on day 0 and day 107	Hapten inducing delayed type hypersensitivity reaction
Deoxycholic acid and polidocanol	NCT06120036	2023	20	Prospective	Clinician assessments of cNFs using a questionnaire and the modified Skindex to assess cNF-related quality of life	Deoxycholic acid destroys adipose tissue polidocanol is a sclerosing agent

Abbreviation: cNF = cutaneous neurofibroma.

High-Frequency Ultrasound

High-frequency ultrasound (HFUS) is a noninvasive imaging technique that uses ultrasound waves to create real-time images of both surface and deeper skin layers within minutes. It can visualize various cNF features, such as shape, borders, and internal characteristics.¹¹ In a prospective study (29 adult patients with 108 cNFs), cNFs appeared hypoechoic, round or ovoid, with well-defined dermal borders.¹¹ A prospective study involving 108 children with NF1 proposed a clinical-HFUS classification system that may aid in developing targeted therapies.²² Two prospective studies^{11,22} showed that HFUS has good to excellent inter-rater reliability (kappa = 0.67–0.95) for assessing surface characteristics, global echogenicity, heterogeneity, and excellent intra-/inter-rater reliability (ICC = 0.97) for volume, width, and depth measurements.¹⁵

Noninvasive Optical Imaging Methods

Li et al.¹⁵ and Jia et al.¹⁴ introduced new noninvasive optical imaging methods, the spatial frequency domain imaging (SFDI) and optical coherence tomography (OCT) for detecting early-stage cNFs beneath the skin. SFDI assesses skin's scattering and absorption, creating maps that can identify potential nascent cNFs by detecting optical differences from surrounding tissue, while high-resolution 3-dimensional OCT images reveal microstructural and vascular changes. Together, they suggest that optical scattering may be a useful marker for early cNF detection and monitoring treatment response.

Wang et al.¹⁶ proposed the use of CT and MRI for evaluating cNFs, but no studies have specifically investigated their application for assessing cNF lesions.

Paper Frames

Cunha et al.¹⁰ evaluated the accuracy of quantifying neurofibromas using a paper frame. A square measuring 100 cm² was applied to 3 specific areas of the body (back, abdomen, and thighs). The average number of neurofibromas per 100 cm² of skin was calculated by averaging 3 values (abdomen, back, and thigh). There was excellent inter-rater reliability, with an ICC range of 0.992–0.997. This methodology was applied in the Cannon et al.¹² study over 8 years to quantify the number and size.

Effect of cNF on HRQoL

We identified 5 studies that specifically evaluated the effect of cNFs on QoL, spanning from 2021 to 2024.^{7,8,23-25} The effect of cNFs on QoL was assessed using various scales and PRO questionnaires (SF36, Skindex 29, Dermatology QoL Index (DLQI), Neurofibromatosis 1-Adult QoL (NF1-AdQOL), and cNF-Skindex) (summarized in eTable 6). Only the cNF-Skindex questionnaire was specifically validated for cNFs (Table 3).

All studies emphasized the importance of PROs in evaluating treatment efficacy and patient-perceived burden.^{7,8,23-25} Maguiness et al.⁷ and Dance et al.⁸ found that greater cNF number and facial involvement were linked to worse QoL, using Skindex-29, DLQI, and NF1-AdQoL. Dance et al.⁸ found higher scores in the emotional domain of the Skindex-29, highlighting the strong emotional effect of cNFs. Lin et al.²⁵ reported that cNF number, size, and facial severity correlated with worse Skindex-29 scores.

The cNF Skindex, derived from the Skindex-16, was designed to assess cNF-related QoL effect with discriminative purposes.²³ Its initial validation demonstrated feasibility and content validity. Reliability was tested via test-retest, but not internal consistency. Construct validity was partially assessed through convergent validity; however, divergent validity and responsiveness were not evaluated.^{23,26} Therefore, comprehensive validation within the NF1 population with cNFs is still needed.

Studies on cNFs Treatments

We identified 23 published studies²⁷⁻⁴⁹ (1987–2024) and 12 ongoing clinical trials on cNF treatments. Of the studies, 16

were prospective and 7 were retrospective; 6 studies focused on topical or oral medical therapies,^{39,40,43,44,47,49} while 17 examined destructive approaches.^{27-38,41,42,45,46,48} All 12 clinical trials were prospective, with 8 focused on novel medical therapies and 4 on destructive techniques comparing 980-nm and 755-nm lasers, radio-frequency injection, and deoxycholic acid injection. Trial statuses included 4 actively recruiting, 2 not recruiting, 4 completed without published results, and 1 with unknown status. Eight out of 12 trials included a physical measurement of cNF quantity or size, while 4/12 trials included an assessment of participant-reported QoL (summarized in eTable 7).

Drug Therapies for cNFs

Imiquimod, a toll-like receptor 7 agonist, modifies innate immune responses. In a completed clinical trial,⁵⁰ a 5% topical imiquimod preparation was tested, with tumor volume as the primary outcome; however, suggested tool was not reported, and the results remain unpublished (Table 2).

Reduced inflammation through nonsteroidal anti-inflammatory drugs was proposed as a potential treatment mechanism.⁵¹ A prospective cohort study involved topical diclofenac gel,⁴³ while Geller et al.³⁹ administered intralesional injections of diclofenac (25 mg/mL) once weekly to 3 target cNFs. In both studies, the outcome assessments involved precise measurements of the 2 evaluated lesions along 2 axes, although the tools were not specified. Standardized digital photography was conducted for both the target and control lesions, with a measuring tape included for reference.

The mTOR pathway is a key regulator of cell growth and Schwann cell survival, and mTOR inhibitors (e.g., sirolimus and everolimus) have been assessed as potential cNF treatments. Kaneda et al.⁴⁹ conducted a pilot RCT study of sirolimus gel (0.2% and 0.4%), with a primary outcome of tumor volume reduction using CT-based volumetric software and ruler-based diameter measurements. A single arm study of everolimus,⁴⁰ assessed its effect in cNF volume reduction, evaluated through photographs and height measurements (no tools were reported).

Table 3 PRO Measures Used for Assessing the cNFs Effect on QoL

PRO	Use	Specifically validated for cNFs
Short Form (SF)-36	Generic scale	No
Skindex 29	Assess physical and psychological effects of skin conditions	No
Dermatology Quality of Life Index	Widely used QoL index for children or adults with diseases affecting the skin	No
NF1-AdQOL	Adults specific QoL measures in NF1	No
cNF-Skindex	Specifically, assess cNF-related QoL in NF1	Yes (content validity, feasibility, reliability, construct validity)

Abbreviations: PRO = patient-reported outcome; QoL = quality of life.

Mitogen-Activated Protein Kinase Kinase (MEK) inhibitors have effectively suppressed Mitogen Activated Protein Kinase signaling, which activates extracellular signal-regulated kinase through phosphorylation.⁵¹ Selumetinib, an oral selective MEK inhibitor, was tested for cNF in a prospective cohort study, with outcomes assessed by calipers and Skindex-29.²⁰ Mirdametinib is a selective MEK1/2 inhibitor, is under investigation in a clinical trial, using 3D photography and digital calipers to track size changes.¹⁹ A topical MEK inhibitor, NFX-179, showed localized Mitogen-Activated Protein Kinase inhibition with minimal systemic effects in a phase I trial; lesion volume was estimated via ruler-based dimensions. A phase II dose-response study is ongoing.⁴⁷

Ranibizumab, an antiangiogenic monoclonal antibody, is being studied in a prospective trial for its effect on tumor vasculature, using calipers and photography for assessment.⁵²

Photodynamic therapy (PDT) is a cancer treatment that involves the selective uptake of a photosensitizer (PS) by tumor cells, followed by light irradiation to activate the PS and induce targeted tumor destruction. PDT was evaluated in a phase I trial using Aminolevulinic Acid as the PS. Lesion dimensions were measured pretreatment and post-treatment, but tools were not specified.⁴⁴ An ongoing clinical trial is testing Levulan Kerastick, a topical PS, in combination with red light PDT, using ultrasound and digital photography to assess lesion regression in adolescents and young adults.⁵³

Diphencyprone (DPCP), a topical hapten that activates the immune system, is under investigation as a cutaneous immunotherapy for cNFs. An open-label phase I trial evaluated its safety and tolerability as primary outcomes, with tumor regression as a secondary end point. Lesions were monitored through serial photographs, including whole-body imaging on days 0 and 107 to assess off-target effects. Although the trial was completed in 2024, results have not yet been published.⁵⁴

Injectable agents such as deoxycholic acid and polidocanol, commonly used in aesthetic medicine, are being explored for cNF treatment. Deoxycholic acid disrupts adipose tissue and may reduce cNF volume, while polidocanol, a sclerosing agent, is being tested for its potential to reduce cNF vascularity and size. An ongoing clinical trial is assessing both agents using clinician-reported questionnaires and a modified Skindex to evaluate QoL effects.⁵⁵

Destructive Techniques

Surgical Resection

Surgical excision targets raised cNFs but is ineffective for flat lesions. The procedure requires complete removal into the deep dermis to minimize recurrence risk and is typically limited to 5 lesions per session.⁴² Onesti et al.³³ introduced a “mega-session” approach, removing a median of 330 large cNFs (>4 cm) in a single operation, with healing by primary or secondary intention based on lesion size. Outcomes were assessed through clinical scar evaluation and patient

satisfaction interviews. A modified biopsy removal can be performed in outpatient settings using standard medical tools. Chamseddin et al. assessed adverse events and patient satisfaction using the DLQI questionnaire and photographic documentation⁴¹ (Table 4).

Electrodesiccation

This method involves dehydrating and denaturing dermal tissue and is favored for addressing numerous flat or superficial lesions in a single session. Patient satisfaction was generally measured using informal verbal surveys.³⁸

Laser

Multiple laser types have been used for cNFs, including CO₂,^{27-30,34,48} Nd: Yttrium Aluminium Garnet (YAG)³² Erbium:YAG,^{36,37} 980-nm diode, and 755-nm alexandrite lasers with suction.⁴⁶ Nine studies (5 prospective and 4 retrospective) have evaluated the efficacy of different laser forms, primarily focusing on superficial or flat lesion removal. Lesion regression and patient satisfaction, often assessed via nonvalidated verbal reports, served as key outcomes. Some studies incorporated 2D/3D photography for before-and-after comparisons. One comparative study³⁶ between CO₂ and Erbium: YAG lasers used the Beausang scar assessment scale to evaluate cosmetic outcomes.

Radiofrequency Ablation Technique

This technique uses a high-temperature electrode to deliver concentrated energy, minimizing damage to surrounding tissue. A retrospective study assessed patient and clinician satisfaction using the Vancouver Scar Scale and the Patient and Observer Scar Assessment Scale.³⁵

High-Intensity Focused Ultrasound

The 20-MHz High-intensity focused ultrasound (HIFU) device provides an acoustic wavelength suitable for selectively treating small lesions within the epidermis or dermis. Peltonen et al.⁴⁵ assessed its safety, feasibility, and efficacy using 3D imaging and patient feedback captured on Likert scale rating 0–10 or 5-point scales. Tumor response was scored from 0 to 10, and ultrasound imaging measured tumor depth and margins pretreatment and post-treatment.

Registered Clinical Trials for Different cNF Interventions

We identified 12 registered clinical trials focused on cNF interventions. Of the 4 trials that appear as completed, 1 has published results; 1 trial has an unknown status. Of the 12 trials identified, 8 are for medical therapies (systemic or topical), and 4 compare different laser modalities and/or topical injections as described below (Table 5).

1. NCT04730583 phase 1 study comparing 1064-nm laser, 755-nm Alexandrite laser, and deoxycholic acid injections. Primary outcome is the incidence of adverse events; secondary outcomes include unspecified patient- and clinician-reported questionnaires.⁵⁶

Table 4 Surgical Resection and Device-Based Techniques

Treatment	Citation	Year	No of patients	Study design	Outcome measures
Surgical excision (Megasection)	Onesti et al. Fijałkowska et al.	2010 2020	15 39	Prospective Retrospective	Clinical scar assessment, patient interview Photographs
Modified biopsy removal	Chamseddin et al.	2019	12	Case series	Dermatology Life Quality Index and photographs
Electrodesiccation	Levine et al. Lutterodt et al.	2008 2016	97 6	Retrospective	Direct questioning at the time of each postoperative visit Telephone survey, verbal report
Erb-YAG laser	Kriechbaumer et al. NCT00921037 Kriechbaumer et al.	2012 2009 2014	12 12 21	Prospective Prospective Prospective	Histologic evaluation Rosenberg's Self-Esteem Scale, SF-36 Photographs, Beausang and VAS Scale
Nd-YAG laser	Elwakil et al.	2008	12	Prospective	Interviews
Comparison of 4 methods Radiofrequency needle coagulation 755-nm alexandrite laser, 980-nm diode laser, intratumoral injection of 10-mg/mL deoxycholate	Richey et al. NCT04730583 NCT06132165	2024 2021 2023	19 20 30	RCT Prospective Prospective	2D, 3D imaging, and objective measurement of height and volume PRO, clinical-reported outcome questionnaires Questionnaire along with 2D and 3D photography
Radiofrequency ablation	Kim et al.	2013	16	Prospective	VSS, POSAS SCALES
HIFU	Peltonen et al.	2024	20	Prospective	Thickness measured by ultrasound, 0–10 scale to evaluate tumor size
CO₂ laser	Roenigk et al. Moreno et al. Algermissen et al. Chiang et al. Meni et al. Peltonen et al.	1987 2001 2001 2012 2015 2022	4 13 119 7 106 256	Retrospective Prospective Retrospective Prospective Retrospective Retrospective	Not mentioned Photographs, questionnaire Patient-reported (unclear method) Telephone survey Survey with unvalidated questionnaire designed for this study 9-question survey

Abbreviations: POSAS = Patient Observer Scar Assessment Scale; SCALES = need to be removed it is referring to the VSS and POSAS scales, already mentioned; VAS = Visual Analog Scale; VSS = Vancouver Scar Scale.

- NCT06132165 will assess the efficacy of skin cooling to improve tolerability across 4 treatments: 980-nm and 755-nm lasers, polidocanol, and deoxycholic acid injections. Outcomes include 2D/3D photography and patient and clinician questionnaires to evaluate lesion appearance and healing.⁵⁷
- NCT00921037 to assess the efficacy of Erbium: YAG lasers for the removal of multiple cNFs. Prospective study including QoL outcomes such as the Nottingham Health Profile, Rosenberg Self-Esteem Scale, and SF-36.⁵⁸
- NCT06300502, not yet recruiting, plans to assess the effectiveness of monthly treatments using a 1064-nm laser, a 755-nm laser, deoxycholic acid, and polidocanol injections.⁵⁹

Discussion

In this scoping review, we identified multiple tools for assessing, quantifying, and evaluating cNFs in treatment studies. While several tools demonstrated good overall reliability, some had no published reliability estimates, and there were notable gaps in other measurement properties. Assessment

of cNFs and their response to treatment was heterogeneous, with no single standardized approach used across studies. This is important, as comparison between studies is difficult when different outcome tools are used. In addition, some methods for quantifying cNFs, such as calipers, are time-consuming, which can limit their feasibility. This is especially relevant for systemic therapies where full body quantification is not possible, and where choosing a single region for measurements may not reflect the overall effect of cNFs in an individual. For physical interventions, where treatments are usually localized to 1 segment per session, quantification of cNFs in the treated area is more feasible.

Although digital calipers are widely used, only 1 study (Gross et al.)²⁰ assessed the minimal detectable difference (MDD), showing a –17.9% mean volume reduction across 126 cNFs after 4 cycles of selumetinib, and a –36.7% average reduction among 5 participants completing 12 cycles (range: –81.4% to –8.3%). While this study examined MDD, no clear estimates exist for the minimal important difference (MID), which relates to whether changes are meaningful for patients. This limits the interpretability of treatment effects. Thalheimer et al.,¹³ suggested that a 25% change in diameter may

Table 5 Registered Clinical Trials for cNF Interventions

Registration	Year	Cohort (prospective or retrospective)	Medical TTT (topical/systemic)	Destructive (laser, surgery, excision, electrodessication)	Completed	Published
NCT00657202	2008	Prospective	Ranibizumab injection		Yes	No
NCT00921037	2009	Prospective		Erbium-YAG-laser	Unknown	No
NCT00865644	2009	Prospective	Imiquimod 5% cream		Yes	No
NCT02728388	2016	Prospective	Levulan Kerastick topical photosensitizer and red-light photodynamic therapy (PDT)		Recruiting	No
NCT02839720^a	2016	Prospective	Selumetinib sulfate (selumetinib)		Yes	Yes
NCT04730583	2021	Prospective	Deoxycholic acid injection	1064-nm laser 755-nm Alexandrite laser	Completed	No
NCT05438290	2022	Prospective	Diphencyprone (DPCP) 0.04% topical ointment once weekly		Yes	No
NCT06132165	2023	Prospective	Polidocanol injection deoxycholic acid injection	980-nm laser 755-nm laser	Recruiting	No
NCT06159166	2023	Prospective	Mirdametinib monotherapy		Recruiting	No
NCT06120036	2023	Prospective	Deoxycholic acid injection polidocanol injection		Recruiting	No
NCT05005845	2023	Prospective	0.05% NFX—179 topical gel (n = 12, 60 cNFs), 0.15% (n = 11, 55 cNFs), 0.5% topical gel (n = 12, 60 cNFs)		Recruiting	No
NCT06300502	2024	Prospective	Deoxycholic acid injection polidocanol injection	1064-nm laser 755-nm laser	Not recruiting	No

Abbreviation: TTT = need to be removed please.

^a Published during the scoping review.

be meaningful, although this remains unvalidated. Cannon et al.⁶⁰ reported that adults with NF1 viewed a 66% reduction in lesion size and number as meaningful, based on a survey. However, we did not find studies using established MID methodologies to define clinically meaningful thresholds for cNF measurement tools.

Many studies have used nonspecific questionnaires that may not fully reflect the distinct effect of cNFs. Evidence consistently links higher lesion counts and facial involvement with worse QoL, underscoring the significant psychological and social burden of cNFs.⁶ Therefore, cNF-Skinindex is a promising, cNF-specific tool, but still needs further validation to assess its reliability and responsiveness, including assessing the MID to interpret changes after an intervention.²³

This lack of validation, including interpretability data for most outcome measures, may explain the variable outcomes used and reported across studies. Older studies focused on surgical and destructive methods, such as excision, electrodessication, and laser ablation, consisted mostly of retrospective studies and case series. These relied mostly on nonspecific or non-validated patient satisfaction surveys. For instance, 1 study evaluated outcomes of mega-session surgery using scar assessments and patient interviews,³³ while another relied on follow-up phone surveys after electrodessication.³⁸ Similarly, studies on CO₂ laser treatments frequently lacked details on the measurement tools or their validation. More recently,

registered studies included RCTs and prospective non-randomized comparative studies. However, there is still variability in the outcomes used, with some trials using validated tools such as digital calipers and 2D/3D imaging, and some reporting changes in lesion size or volume without detailing the measurement tools, procedures, or criteria used to interpret results. These inconsistencies make it hard to assess if a given intervention provides clinically meaningful benefits and prevent comparison across modalities and trials. These gaps underscore the need for further work on validated, condition-specific instruments such as the cNF-Skinindex, to enable more rigorous evaluation of treatment outcomes.

We also identified several completed trials with unpublished results. This raises concerns about potential reporting bias, especially if negative or inconclusive findings are unpublished, further limiting the knowledge base.

In summary, we found multiple gaps in the available measures aimed at quantifying cNFs, their effect on QoL, and response to treatment. Limitations in outcome measurement validation directly affect clinical trial development and interpretation, subsequently affecting patient care. The cNF Skinindex shows promise for assessing the effect of cNFs in QoL but needs further validation. There is a pressing need to standardize cNF quantification methods, assess their responsiveness, and determine MIDs. These findings support the ongoing efforts of the REiNS (Response Evaluation in Neurofibromatosis and

Schwannomatosis) cNF subgroup^{13,60} to develop core outcomes and reliable measurement tools for cNFs research.

Author Contributions

H.A. Elahmar: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. C.A. Soto Rincon: major role in the acquisition of data. A. Drucker: drafting/revision of the manuscript for content, including medical writing for content. P. Parkin: drafting/revision of the manuscript for content, including medical writing for content. C. Barnett-Tapia: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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